

(PS6) Q2: $B \cdot m = e^{m-L} = \left(-\frac{Am_f}{m} \right) \cdot m$

* $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$

Most cases:

$$\langle f, g \rangle = \int f(x)g(x) dx$$

* More generally: weighted inner-product:

$$\langle \cdot, \cdot \rangle_w$$

$$\langle f, g \rangle_w = \int f(x)g(x) \underbrace{w(x)}_{\text{weight}} dx$$

S.L. theory:

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x)y = -\lambda \underbrace{w(x)}_{\text{weight}} y$$

* (b) $Y_n(x) = \begin{cases} \dots & n=0 \\ \sqrt{\dots} e^{-x} \begin{bmatrix} \dots \end{bmatrix} & n \geq 1 \end{cases}$

(c) $c_n = \langle \underbrace{e^{-x}}_{f(x)}, Y_n \rangle_w \leftarrow$

$f(x) = e^{-x}, \quad w(x) = e^{2x}$

$$c_n = \int e^{-x} \cdot Y_n(x) w(x) dx$$

$$= \int e^{-x} \sqrt{\dots} e^{-x} \begin{bmatrix} \dots \end{bmatrix} e^{2x} dx$$

$$= \int \underbrace{e^{-2x} \cdot e^{2x}}_1 \sqrt{\dots} \begin{bmatrix} \dots \end{bmatrix} dx$$

(3) Assume $\lambda = 0 \Rightarrow$ contradiction.

$\hookrightarrow * \underline{X}'' = -\lambda \underline{X} *$

$\vdots$

$\sqrt{\lambda}$

$\hookrightarrow$ change to λ^2

(4) Run = DDE: Start as usual or as interval

$$u(x,t) = \sum_{n=0}^{\infty} b_n(t) Y_n(x)$$

↓ once you get to the ODE for X_n is when we use later methods:

→ Sep. of var. $u(x,t) = T(t) X(x)$

• plug in:

$$X'' = -\lambda^2 X$$

← ODE S.L.

$$w(x) = 1$$

→ Find solⁿ X_n, T_n , plug in:

$$u(t,x) = \sum C_n X_n e^{-\lambda_n^2 t}$$

$$\text{I.C. } u(0,x) = f(x)$$

• To find C_n :

*

$$C_n = \langle f(x), X_n(x) \rangle$$

↳

$$\int f(x) X_n(x) dx$$

$$(7) \quad r^2 R'' + r R' = -\lambda R, \quad R(1) = R(p) = 0$$

$$\lambda = 0: R(r) = A + B \log(r)$$

$$R(1) = A + B \log(1) = 0$$

